# Tradable services and local job multipliers: evidence from Brazil

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**Abstract**

The rapid growth of cross-border trade in services, enabled by digitalization and falling trade costs, has transformed the nature of globalization. Despite this transformation, little is known about the local labour market effects of employment growth in tradable service activities. This paper sheds new light on how expansions in tradable services employment spill over onto jobs in non-tradable services. Drawing on matched employer–employee administrative microdata covering the universe of formal workers in Brazil between 1995 and 2023, we construct a granular classification of tradable and non-tradable services based on geographic concentration. We then estimate local job multipliers using a panel shift-share instrumental variable design that exploits growth in tradable services subsectors, building upon traditional methodologies. Our results indicate that tradable services generate substantial local spillovers: for every additional job in tradable services, approximately 1.3 jobs are created in non-tradable services. This effect is larger than that of the overall tradable sector, including manufacturing. Additional analyses suggest that increased household demand is the primary mechanism driving these spillovers, which tend to be stronger for younger and male workers and more pronounced in larger local labour markets. While our baseline empirical design does not directly observe trade flows, the findings are consistent with the growing importance of service activities that can serve non-local demand in shaping local labour market dynamics. These results highlight the need to incorporate tradable services into discussions of regional development and employment creation.

**Keywords:** Job multipliers, local labour markets, tradable services

**JEL classification:** F16, R12, R23

# 1 Introduction

Over the past two decades, global trade in services has grown rapidly, supported by the diffusion of digital technologies that have reduced the cost of delivering services across distance (Benz, 2022; Herman, 2023; WTO, 2019). These developments have expanded the set of services that can be supplied beyond local markets, blurring the traditional distinction between tradable and non-tradable activities. Despite this transformation, much of the academic and policy debate on the labour market effects of trade continues to focus on manufacturing and to treat services as predominantly local, non-tradable activities.

This characterization is increasingly at odds with both theory and evidence. A growing share of services — particularly in finance, business services, and information technology — can now serve non-local demand, whether across regions within a country or across borders. Yet empirical research on services trade faces two major challenges. First, data limitations make it difficult to directly measure trade in services at a granular level. Second, there is no consensus in the literature on how to classify services by tradability, limiting the scope for systematic analysis of their local labour market effects.

This paper studies the local employment spillovers generated by employment growth in tradable services sectors. In particular, it asks whether expansions in these sectors — which are capable of serving non-local demand — generate additional employment in local non-tradable activities. The quantification of such spillovers is commonly referred to as the job multiplier, defined as the number of additional jobs created in the non-tradable sector for each new job in the tradable sector in a local labour market.[1]

Modern empirical literature on job multipliers builds on the seminal study by Moretti (2010), who uses a Bartik (1991) shift-share instrument to measure how many additional jobs in non-tradable sectors were created for every new job in manufacturing in US metropolitan statistical areas. Since then, most studies have continued to focus on manufacturing job multiplier effects, providing evidence that new employment is created as a result of job growth in the tradable sector in the United States (Gerolimetto and Magrini, 2016; Nguyen and Soh, 2017), Europe (Cerqua and Pellegrini, 2020; de Blasio and Menon, 2011; Goos et al., 2018), China (Wang and Chanda, 2018) and Japan (Kazekami, 2017). Recent studies have acknowledged the importance of tradable

[1] Multiple terms can be used to refer to local labour markets, which can vary by country of analysis. For the purpose of this study, local labour markets will also be referred to as regions or, empirically, as microregions.

services and tried to integrate them in a pooled traded sector with manufacturing (Frocrain and Giraud, 2019; Osman and Kemeny, 2022; Van Dijk, 2018). In a rare exception, Avdiu et al. (2022) examine the job multiplier effects exclusively of tradable services in the case of India.

Focusing on Brazil, this study combines detailed rich administrative data with recent methodological advances to analyse the job multiplier associated with tradable services. Using matched employer–employee data covering the universe of Brazilian formal employment, we construct a granular classification of tradable and non-tradable services and exploit variation in employment growth across local labour markets. Methodologically, the paper builds on the established job multiplier framework while leveraging high-frequency panel data and an adapted shift-share instrumental variable design to isolate variation in tradable services employment driven by non-local demand. As Bartik and Sotherland (2019) show and Osman and Kemeny (2022) discuss, while most studies have to rely on a single cross-section difference of census data, such estimates tend to be sensitive to start-end period arbitrary choices and have less statistical power to control for temporal shocks and local labour market idiosyncracies. Borusyak et al. (2025) also highlight that having multiple periods in the specification can increase the estimation power of the shift-share instrumental variable, nearly universally used in the job multiplier literature. Therefore, this paper contributes to the literature by applying an adapted SSIV panel design to study the job multiplier effects of tradable services specifically, while also providing evidence for Brazil.

Our findings suggest that, for every new job in tradable services, 1.3 new jobs in non-tradable services are created in Brazil, at least in the short-term. This estimate is higher than the job multiplier for the total tradable sector and at par with other findings in the literature. These effects tend to be stronger for men and younger workers, and the main mechanism driving job growth is increased household demand. We also find that, while larger microregions have stronger job multipliers, our results are not exclusively driven by large cities like São Paulo. Finally, results suggest that spillovers are not exclusively about employment, and that hours worked and wages of workers in non-tradable services also increase thanks to job rise in tradable services employment.

The remaining part of this paper is organized as follows. Section 2 reviews some theoretical concepts and the current empirical literature on trade and labour markets. Section 3 outlines data sources and estimations, as well as descriptive statistics on tradable and non-tradable services and the final sample. Section 4 describes the empirical strategy, and section 5 presents results. Section 6 concludes.

## 2 Conceptual framework

Job multiplier analysis is inserted in the broader literature of the impact of trade on labour markets. Most studies in this field focus on direct effects of trade exposure on workers in the exposed sector or region. The seminal Autor, Dorn and Hanson (2013) paper highlights a common researhch question, which is trying to estimate the impact of import competition - the China shock - on manufacturing workers. Studies in the field broadly agree on the persistent employment and wage losses in exposed sectors and regions (Dix-Carneiro and Kovak, 2017; Malgouyres, 2017). More recently, important nuances to such movements have been made to account for the significant portions of workers that reallocated from manufacturing to services in developed economies (Bloom et al., 2024). On a different angle of trade exposure, increased export opportunities are found to translate into increased worker's wage and within-sector firm switching (Dauth et al., 2021).

Job multiplier analysis differs from these studies by focusing on spillover effects on workers not directly exposed to trade. Theoretically, job multipliers should be interpreted within a spatial general equilibrium framework (Moretti, 2010). In this setting, the economy consists of multiple regions (equivalent to local labour markets) producing two types of goods and services: tradables, whose prices are determined in external markets, and non-tradables, whose prices and quantities are determined locally. Regions are populated by mobile workers who choose locations based on wages, prices, and idiosyncratic location preferences, implying an elastic local labour supply. Housing supply is also elastic but constrained by local land-use regulations and geographic characteristics, which affect the extent to which increases in labour demand translate into population growth rather than higher local prices.

An exogenous increase in demand for a region's tradable output — such as a rise in foreign demand for a tradable service — raises labour demand in the tradable sector. This shock leads to a direct increase in local employment in a given region, which may be met through migration to the region, increased labour force participation, or reallocation across sectors within the region. To the extent that the employment expansion is not fully offset by sectoral reallocation, net local employment and regional household income increase.

Beyond this immediate rise in local employment in the tradable sector, there are indirect spillover effects in the non-tradable sector, commonly referred to as job multiplier effects. A first mechanism operates through the income or consumption channel: as employment and wages in the

tradable sector rise, higher household income increases demand for locally produced services, such as personal care, retail, and hospitality. A second mechanism reflects local input-output linkages, whereby expanding tradable firms increase demand for locally supplied intermediate services, such as security or maintenance. A third channel operates through agglomeration and productivity effects, as increased spatial concentration can raise productivity and employment across sectors (Beaudry and Schiffauerova, 2009).

As highlighted by Moretti and Thulin (2013), the magnitude of the multiplier depends on the strength of these channels. For instance, the extent to which higher household income translates into additional non-tradable employment depends on wage levels in the tradable sector and on local consumption patterns. At the same time, general equilibrium adjustments can attenuate multiplier effects. A constrained housing supply may lead to rising living costs, limiting in-migration and pricing low-wage workers out of the region. Similarly, tight labour markets — due to low unemployment or limited worker mobility — may cause increases in demand for the non-tradable sector to be absorbed through higher wages rather than additional employment.

Many studies have tried to quantify the job multiplier in different settings, oftentimes only taking the manufacturing sector as the tradable sector. Moretti (2010) initially found a manufacturing job multiplier of 1.6 in US metropolitan areas, rising as far as 4.9 to high-technology manufacturing. Since, methodological improvements have been added to the framework, pointing to a potential overestimation from this first exercise. For example, excluding the own metropolitan area in the calculation of the SSIV (leave-one-out approach); and using metropolitan areas where people work, rather than live. More recently, analysing US commuting zones, Bartik and Sotherland (2019) arrive at a manufacturing multiplier of 0.1-0.8, and a high-techonology manufacturing multiplier of 0.9-1.0. Closer to our analysis, Van Dijk (2018) estimates a job multiplier of the whole US tradable sector (tradable services and manufacturing) between 0.17-0.88. Frocrain and Giraud (2019) estimate a similar figure of 0.64 for the whole of the French tradable sector. Methodologically closer to our approach, Osman and Kemeny (2022) report a job multiplier of 0.86 in the US tradable sector. Finally, Avdiu et al. (2022), specifically analysing tradable services in India, find that a 10% increase in tradable services employment leads to a 4.2% increase in non-tradable services employment.[2]

[2]In their study, only elasticities are reported. The estimation also differs methodologically from most of the literature, as they approach the shift aspect of the shift-share instrumental variable through services trade data, rather than employment data for tradable services.

# 3 Data

Our primary data source is the Relação Anual de Informações Sociais (RAIS), an administrative dataset collected annually since 1986 by the Brazilian Ministry of Labour. RAIS functions as a census of the Brazilian formal labour market, as all registered firms are legally required to report their workforce in order for employees to access social security, severance pay, and other benefits. Non-compliance can result in fines, which creates strong incentives for accurate reporting by both firms and workers. This institutional framework ensures the reliability of the database and explains why RAIS has become the main source for labour market studies in Brazil (Dix-Carneiro and Kovak, 2017; Naercio A. Menezes-Filho, 2011).

RAIS covers all formal employees in Brazil, i.e. workers with a signed work card (*carteira de trabalho assinada*), which guarantees access to the benefits of the country's labour legislation. The dataset excludes informal workers, self-employed individuals, domestic workers (with some exceptions), interns, and other minor employment categories. Each observation corresponds to an employer–employee relationship in a given year, and a single worker may appear multiple times if employed by different firms. We measure employment using only relationships active on December 31, which yields an annual stock of formal employment consistent with official published figures and avoids double counting. While RAIS was established in Brazil in 1975, the information it gathers has increased in breadth across time. Namely, industries started being reported beyond 25 sectors after 1994, which is why we work with RAIS from 1994 through 2023 at the ISIC Rev. 2-digit level, totalling 88 unique sectors, 51 of which in services.

Because of its administrative nature, RAIS also allows for the observation of employment at fine geographic levels, for over 5,500 municipalities, which we aggregate to 510 microregions (local labour markets) with stable boundaries over time using IBGE's official crosswalk. Microregions have been typically used in the literature as proxies for local labour markets, since they provide a balance between geographic granularity and the size required to capture the economically contiguous concept of local labour markets. In addition, RAIS provides rich sociodemographic information on workers, enabling heterogeneity analyses beyond aggregate employment. At the same time, it is representative at all levels, since it accounts for the whole population of employees. Crucially, RAIS enables the observation of employment at a high-frequency, across all years of the dataset.[3]

[3] As discussed, many studies in the job multiplier literature rely on cross-sectional data, oftentimes census data, that is either infrequent by design, or not robust within short time spans. For example, in France or the United States, census data is collected in a rolling basis, and yearly data also groups information from previous and following

The most significant limitation of RAIS is its exclusion of informal workers, who represent a substantial share of total employment in Brazil. Estimates suggest that between 40% and 50% of workers were informally employed during the 2010s, depending on the data source (ILOSTAT, IPEA). Informality is especially prevalent in agriculture, construction, and personal services, while manufacturing and public sector employment are relatively formalized. For example, in 2019, informality rates exceeded 75% in agriculture and over 50% in several non-tradable service industries, but were close to 8% in manufacturing overall.

These patterns are important when interpreting results, as our estimates capture multipliers in the formal sector, and therefore could be reflecting workers moving from the informal sector into the formal sector, rather than new jobs being created. To control for this, we make use of the Brazilian decennial Census (IBGE) which includes both formal and informal workers. While it is infrequent and less detailed by nature, it allows for a holistic view of the Brazilian labour market.

### 3.1 Defining tradable services

To measure job multiplier effects, we categorise 2-digit service industries into tradable and non-tradable services. In the context of this study, tradable services are services that can meet non-local demand, i.e. beyond the demand of their local labour market. For example, an architect can email plans to a client abroad, while a hairdresser cannot cut a client's hair from afar. This implies that the analysis defines tradable services as those that are mostly deliverable remotely, corresponding to Mode 1 in the WTO context[4] – while also considering within-country trade flows.

Technological progress has lowered the cost of delivering services remotely, increasing their tradability and enabling not only cross-border trade but also domestic exchanges across different regions of a country. Therefore, the definition of tradable services in this analysis encompasses cross-border trade as well as intra-national flows between local labour markets. For example, a service that can be delivered from São Paulo to Rio de Janeiro is considered tradable.

Within this concept, the classification of tradable services in this study builds on Jensen and Kletzer (2005)[5], and measures the geographic concentration of a given sector's (ISIC Rev. 4 2-digit)

years. Which is why comparing employment trends from one year to the other can be highly misleading.

[4]The WTO categorizes trade in services within four modes of supply: Mode 1, where supplier and consumer remain in their respective territories and similar to traditional trade in goods; Mode 2, where the consumer goes to the supplier, commonly reflecting tourist expenditures; Mode 3, where the supplier goes to the consumer through commercial presence, reflecting foreign direct investment and foreign affiliates; and Mode 4, where the supplier goes to the consumer through the presence of natural persons.

[5]The exercise of identifying which services are tradable has been extensively discussed in the literature, that

employment in a given microregion through the following formula:

$$geoconc_i = 100 * \sum_c (S_{ci} - S_c)^2 \tag{1}$$

Where if an industry is highly concentrated in a few microregions beyond their relative share of national employment, $geoconc_i$ increases and the industry is considered concentrated.

Following Frocrain and Giraud (2019), we define the cutoff for tradability as the lowest geographic concentration in a manufacturing 2-digit sector. This definition is calculated for all years of the dataset, and sectors whose definition is not constant over time are defined manually through the analysis of sectorial description in ISIC Rev. 4.

Under this classification, tradable services include the following ISIC Rev. 4 sections: information and communication, finance and insurance, real estate fully; and transportation and storage, science and technology and administrative services partially. Non-tradable services fully include construction; wholesale and retail; accommodation and food; public administration, education, and health; arts, and entertainment; and other service activities.[6]

### 3.2 Descriptive statistics

Table 1 showcases some differences between tradable and non-tradable services in the Brazilian formal labour market. Firstly, there are significantly more Brazilian employees in non-tradable services (NTS), at approximately 35 million compared to 10 million in tradable services (TS). However, this gap has decreased over time, since growth in TS has outpaced that of NTS, especially since 2010. The reasons for a significantly higher relative growth rate in TS since 2010 are twofold: firstly, it reflects the rise in tradability of these services, and therefore of services trade; additionally, it excludes the early 2000s, during which took place a large movement of workers - particularly in NTS - from the informal to the formal sector (Berg, 2010).[7] The exclusion of informal workers also likely leads to an overestimation of the percentage of workers with a college degree, particularly in

generally derives tradability from how instensely service sectors are traded Avdiu et al., 2022; Calligaris et al., 2024; the geographical concentration of production and consumption of services (Frocrain and Giraud, 2019; Jensen and Kletzer, 2005; Van Dijk, 2018); or from a conceptual analysis based on sectorial descriptions (WTO, 2023).

[6] A full breakdown of 2-digit ISIC Rev.4 sectors by geographic concentration and category is presented in Table A.1.

[7] While this change has been associated with the Brazilian economic growth during the 2000s commodity boom, it is also a results of policy. For instance, Abras et al. (2018) show that, in early 2000s Brazil, increased enforcement of labour market regulations in cities led to an increase in gross and net formal job creation.

NTS. In terms of other workforce characteristics, TS tends to employ younger workers and is more male-intensive than NTS. Crucially, as expected from our conceptual and empirical approach to categorising services, TS tend to be significantly more concentrated in the Brazilian territory.

Table 1: Tradable and non-tradable services in the Brazilian formal labour market

| Variable | TS | NTS |
|---|---|---|
| Million employees in 2023 | 9.6 | 34.7 |
| Employment growth between 1995/2010 and 2023 | 177%/54% | 153%/23% |
| % of over 45-year-olds | 26% | 34% |
| % of female workers | 41% | 50% |
| % of workers with college degree | 25% | 25% |
| Geographic concentration (HHI * 100) | 74 | 30 |

Table 2 reports the final sample in the analysis, which is composed of a panel of 510 microregions across 29 years (1995-2023). As expected, microregions differ significantly in terms of size. With a population largely concentrated in coastal areas, Brazil's microregions have a median employment of 63 thousand workers, with a mean of nearly 200 thousand workers. Large local labour markets, such as those centered around São Paulo or Rio de Janeiro, signficantly skew the distribution of employment in Brazil. This is particularly true for tradable services employment. Additionally, while the average yearly TS employment growth rate stood at 6%, outpacing that of NTS and total employment growth, the exogenous (shift-share instrumental variable, SSIV) TS employment growth rate is 3%. This motivates the need for purging certain factors that might contaminate the variation explored in this study, namely the growth in tradable services employment driven by non-local demand.

## 4 Empirical Strategy

The empirical strategy to estimate multiplier effects is based on the following equation:

$$ln(e_{i,t}^{NTS}) = \beta_0 + \beta_1 ln(e_{i,t}^{TS}) + \mu_i + \eta_t + \varepsilon_{i,t} \tag{2}$$

This baseline estimation captures job multiplier effects in the short term. Given the general equilibrium setting, the specification aims to quantify the effects in a broad, reduced-form approach.

Table 2: Sample statistics (510 microregions, all years pooled)

| Variable | Mean | Median | SD |
|---|---|---|---|
| Total employment (incl. agriculture and mining), thousands | 195.8 | 63.1 | 725.5 |
| Tradable employment, thousands | 38.2 | 12.8 | 158.6 |
| Tradable services (TS) employment, thousands | 18.8 | 3 | 113.5 |
| Non-tradable services (NTS) employment, thousands | 69.1 | 20.9 | 262.9 |
| Avg yearly employment growth rate | 5.07 | 4.45 | 2.85 |
| Avg yearly NTS employment growth rate | 5.48 | 4.70 | 2.98 |
| Avg yearly TS employment growth rate | 6.11 | 5.90 | 2.67 |
| Avg yearly TS employment growth rate (SSIV) | 3.04 | 3.18 | 0.90 |
| Avg NTS real hourly wage | 13.31 | 13.20 | 2.12 |
| Percentage of informal workers (Census 1991) | 73.6 | 76.8 | 15.7 |
| Percentage of informal workers (Census 2010) | 62.0 | 60.5 | 14.9 |

That is, to explain levels of non-tradable services (NTS) employment with levels of tradable services (TS). The outcome variable $ln(e_{i,t}^{NTS})$ reflects the log NTS employment in microregion $i$ at time $t$, and $ln(e_{i,t}^{TS})$ represents the equivalent variable for TS. Additionally, the equation contains microregion fixed effects ($\mu_i$), accounting for unobserved constant local characteristics of these local labour markets, like consumer preferences; and year fixed effects ($\eta_t$) to control for time-varying nation-wide shocks in the economy. Therefore, our identification comes from within-region changes in employment, where the coefficient of interest ($\beta_1$) reports an elasticity. The job multiplier is derived by multiplying this elasticity by the average NTS-to-TS employment ratio across all periods.

This specification draws largely from Osman and Kemeny (2022). As in their identification strategy, this study relies on frequent data to follow employment levels across time, instead of long-difference specifications. This allows for a more robust implementation of the specification by enabling a better distinction between signal and noise in the coefficient of interest, which becomes less sensitive to arbitrary choices or constraints dictating the first and last years of the analysis. A longer panel also enables the inclusion of robust microregion fixed effects, which is typically not included in specifications partly due to the issue of shorter panels (Bartik and Sotherland, 2019).

Endogeneity concerns can arise in the OLS estimates. Firstly, unobserved evolving microregion characteristics can be simultaneously affecting both NTS and TS employment, such as shifting crime rates or the effects of education policies. Crucially, tradable services are not always traded: a considerable share of employment in TS is driven by local demand. For instance, while bankers

can meet non-local demand, this does not mean they exclusively do so; many jobs in banking cater to city residents. These concerns motivate the need of capturing variation reflecting exogenous, non-locally driven TS employment growth, as introduced in table 2.

Following most of the job multiplier literature, this analysis relies on a shift-share instrumental variable (SSIV) to do so. Its classic form takes stock of nation-wide shifts (second half of the equation below, reporting a growth rate) in subsectors (2-digit) of tradable services employment, thus exogenous to specific local demand, and attributes them to microregions based on a weight, or share (the first half of the equation below), of how large those subsectors were - relative to overall TS employment - prior to the shift in a given microregion (see Figure 1).

$$SSIV_{i,t}^{TS} = \sum_{j \in TS}^{J} \frac{e_{i,j,t-1}}{e_{i,J,t-1}} * \left\{ \frac{(E_{j,t} - e_{i,j,t}) - (E_{j,t-1} - e_{i,j,t-1})}{(E_{j,t-1} - e_{i,j,t-1})} \right\} \tag{3}$$

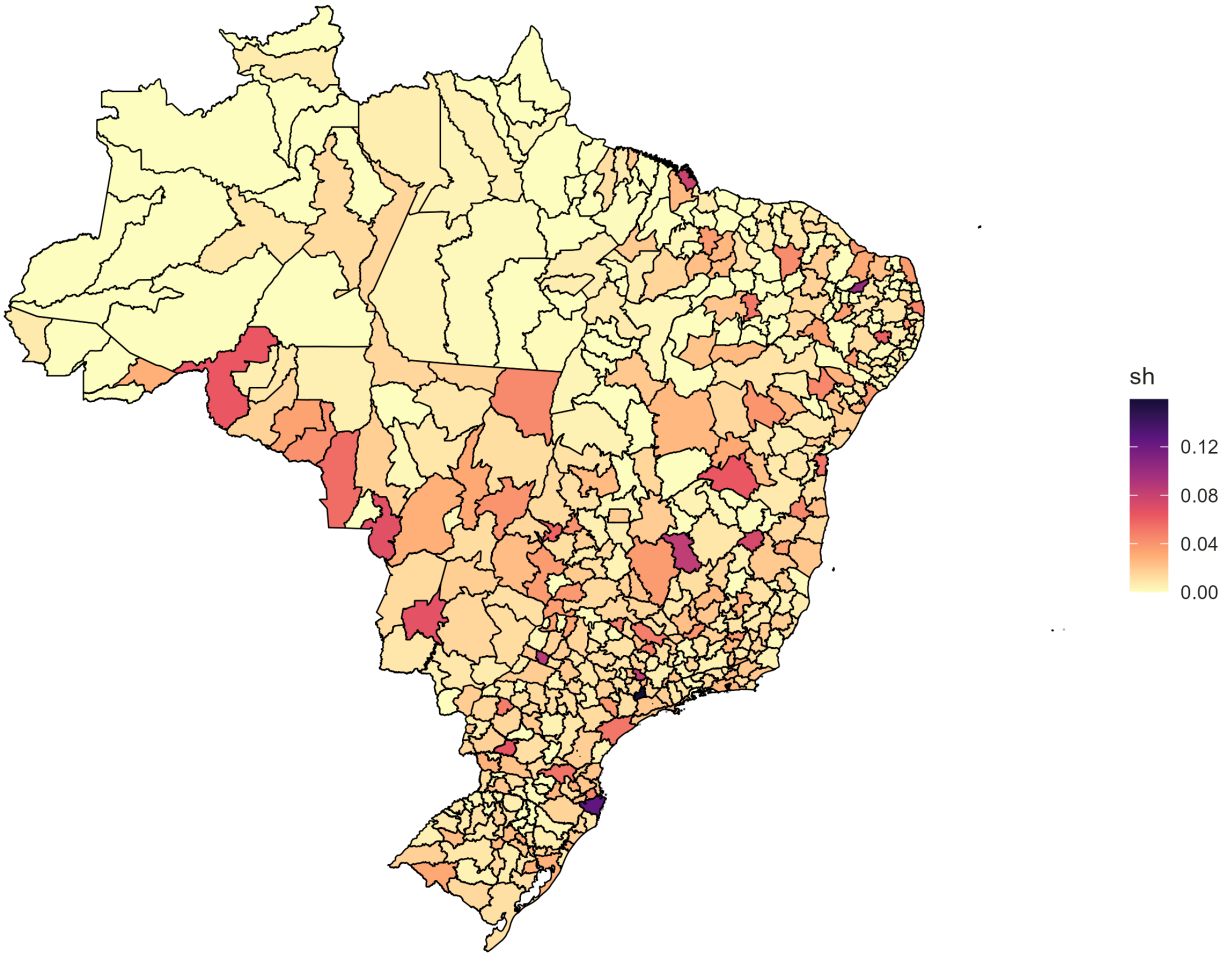
Where $j$ is a 2-digit subsector of tradable services $J$; $t-1$ and $t$ report the previous and current years of the analysis for a given microregion $i$. Following Borusyak et al. (2025), a leave-one-out approach is adopted. That is, for a given microregion, the nation-wide shock is the country's shift (($E_{j,t}$) minus its own shift ($e_{i,j,t}$). This ensures that microregions with a large weight of employment in a given sector cannot influence their allocated exogenous, non-local shift, and is particularly important for sectors that are geographically concentrated, such as tradable services.

The strength of the quasi-national shift as a predictor of a microregion's TS employment growth precludes the idea that these services are indeed tradable - that competition, prices and demand operate to some degree at a national and international scale. In other words, misclassifying NTS as TS would severely weaken the instrument. If NTS are by definition responding to mostly independent demand trends of each local labour market, the aggregate shift of NTS employment growth would hardly inform on a given microregion's own NTS employment growth.

A key choice in this SSIV specification concerns the definition of pre-exposure shares. While many studies in the job multiplier initially relied on a single long-difference design, now most studies work with stacked differences or, in our case, panel specifications. In the case of multiple periods of analysis, Borusyak et al. (2025) that the exposure from shares in the SSIV can be fixed at the base period or be lagged to account for re-specialisation. Given the long span of our panel (1995-2023) and the substantial structural transformation of the services sector in Brazil over this period, we opt to lag shares, in line with Osman and Kemeny (2022). The underlying intuition

is that the composition of tradable services has significantly shifted between 1995 and 2023, away from financial services, real estate and land transport and into administrative office, employment and computer programming activities.[8] Therefore, relying on on base period (1995) shares to define a microregion's sectoral composition would likely be misrepresentative of its exposure to TS growth in later years due to such structural changes. At the same time, using time-varying shares raises the possibility of endogenous re-specialization, whereby local shocks may influence sectoral composition. We will aim to test for such threats to the specification with alternative specifications.

Figure 1: Illustrating shares in the SSIV: The weight of IT services in Brazilian microregions, 2008




Source: Relações Anuais de Informações Sociais, Ministry of Labour.

[8]Table A.2 reports the changes in the 2-digit sector composition of tradable services in Brazil between 1995 and 2023.

# 5 Results

## 5.1 Job multipliers: the size of job creation in non-tradable services led by job growth in tradable services

Following the identification strategy previously described, we now present estimates for our main variables of interest. Table 3 reports how changes in employment in the tradable sector relate to changes in employment in the non-tradable sector. Since this study aims to focus at a subset of the tradable sector (tradable services), results are compared to effects derived from the overall tradable sector (including manufacturing). Such approach benchmarks our findings to the recent job multiplier literature.

Models 1 and 2 report the OLS results, showing a significant and strong correlation between the rise of jobs in the tradable sector, both including (T) and excluding manufacturing (TS), and jobs in non-tradable services (NTS). Models 1 (and 3) suggest that a 10% increase in TS (T) jobs is associated with a 2% (2.4%) in NTS jobs. In terms of job multipliers, since TS is smaller than T by definition, the effect is stronger for TS. For every one new additional job in TS (T), the model predicts 0.9 (0.5) new additional jobs in non-tradable services.

Our shift-share instrumental variable (SSIV) results, based on two-stage least-square regressions, suggest that the the effects are uniformly larger than their OLS counterparts. Strong instruments reinforce the idea that TS are correctly identified, also reflecting the advantage of high-frequency data.[9] In this preferred specification, we also find that the job multiplier of TS is stronger (1.3) than that of T (0.7). These results suggest that strong spillovers of tradable services employment growth on non-tradable services employment should be expected, in particular as employment in tradable services further grows through increased services tradability and services trade liberalization (WTO, 2019).

As discussed, given its general equilibrium framework, the job multiplier is estimated from a reduced-form approach. However, alternative specifications with controls tend to reinforce that job multiplier effects are strong and significant. Table A.4 reports results with a rich set of base period observables, similar to the exercise of Van Dijk (2018): log employment and population, as

[9] A common challenge in the literature, partly driven by low-frequency data, is identifying strong shift-share instrumental variables. Multiple studies report difficulties in dealing with first-stage F-statistics around or below 10 (Moretti, 2010; Van Dijk, 2018)

Table 3: Estimate of the impact of employment in the tradable sector on employment in the non-tradable sector in Brazil

| | *emp.NTS* | | | |
|---|---|---|---|---|
| | (1) OLS | (2) SSIV | (3) OLS | (4) SSIV |
| *emp.TS* | 0.202*** | 0.294** | | |
| | (0.031) | (0.120) | | |
| *emp.T* | | | 0.240*** | 0.354* |
| | | | (0.041) | (0.196) |
| F-stat | – | 314.49 | – | 79.64 |
| Adj. $R^2$ | 0.967 | 0.966 | 0.962 | 0.940 |
| Observations | 14790 | 14790 | 14790 | 14790 |
| FEs | geo, yr | geo, yr | geo, yr | geo, yr |

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. SEs clustered at microregion level.

well as share of informal workers[10], female workers, workers with college education and workers in manufacturing. Column 4 reports SSIV results with no microregion fixed effects, and column 8 reports SSIV results with state-level (geographic aggregates) fixed effects. While controls are highly significant and help explain where NTS job growth happened in Brazil, they do not weaken the causal link of the job multiplier. Most importantly, results in columns 4 and 8 suggest a similar magnitude of the effect: 1.4 and 1.3 new NTS jobs for every new TS job, respectively.

A subsequent alternative specification with controls entails the usage of moving observables. Table A.5 compares the baseline results of table 3 with models 3-4, reporting additional controls. These models further show a stable coefficient for the job multiplier. NTS job growth is positively correlated with the share of female and college-educated workers and negatively correlated with the share of manufacturing workers. However, incluing these covariates does not affect the magnitude of the main covariate, which still points to a 1.3 job multiplier between TS and NTS jobs.

Taken together, these results help analyse the relationship between the OLS and SSIV estimate of the job multiplier. In the case of the baseline specification presented in table 3, SSIV coefficients are consistently larger than their OLS counterparts. As previously highlighted, a key feature of tradable services is that they still significantly cater to local demand, which is captured in the OLS estimates. This can both raise an attenuation or upward bias: for instance, in the case of weaker OLS coefficients in the baseline specification, this may signal that TS employment is substantially reliant on local demand and is being partly depressed by microregions with weak local demand. On the other hand, equivalent specifications without microregion fixed effects reported in table A.4

[10]Since both population and size of the informal market are based on Brazilian Census data, the base period must be set at 1991 instead of 1995.

(columns 1-2 and 5-6) point to an upward bias. This may be reflecting how we allow for more co-movement of TS and NTS employment through time-invariant microregion characteristics, such as institutional settings that foster employment. This, in turn, leads to an overestimation of job multiplier effects.

### 5.2 Mechanisms behind the job multiplier effect in Brazil

Job multipliers can be driven by different cohorts and mechanisms. To compare heterogeneous effects of the job multipliers, we focus on the reported elasticities. We first turn to a decomposition of TS employment by skill level based on whether workers have a college degree. The literature (Moretti, 2010; Van Dijk, 2018) suggests that growth of high-skilled employment in the tradable sector should be the main driver of job multiplier effects, given its correlation with higher wages. We find evidence challenging this idea. Table A.6 suggests a slightly larger - although not statistically significant - job multiplier effect from low-skilled workers. Despite arguably lower wages, a possibly larger marginal propensity of consumption could be at play to strengthen the nexus of the consumption channel. Crucially, both blue collar workers are also drivers of spillover effects, and policymaking should not underestimate the potential of low-skilled employment as a source of job creation.

A follow-up exercise is to understand if households are really the main drivers of the job multiplier, since intersectoral firm-to-firm linkages could also create such spillovers. Based on the OECD Inter Country Input Output tables, we categorise NTS sectors based on whether they cater mainly to household final demand (B2C, business to customer) or firm intermediate consumption (B2B, business to business).[11] In Table A.7, such decomposition suggests that TS employment growth is highly endogenous to B2B-intensive NTS employment growth, as it loses significance between the OLS and SSIV models. We therefore find that household income, or the consumption channel, is the main driver of the job multiplier in Brazil.

[11] This exercise is done through observing how much of a sector's output is sold to either intermediate consumption (other firms) or household final demand. If a sector is sells more than half of its output to the former, it is considered B2B-intensive; otherwise, it is considered B2C-intensive. An importance nuance is that, for the purpose of these estimates, total output excludes gross fixed capital formation (GFCF), which in the case of services can reflect intangible assets such as software or intellectual property. This is motivated by the fact that GFCF is typically reported as a component of final demand in input-output modelling, which can lead to strong underestimation of firm-to-firm sales. Final categories by sector are reported in Table A.3

### 5.3 Who benefits most from tradable services job multipliers in Brazil?

Taking stock of the baseline results of the job multiplier in Brazil, we decompose NTS employment to shed light on who wins most from these spillover effects. As discussed, distributional impacts are expected as sectoral workforce composition and consumer behavior of Brazilian households and firms can differ from other countries.

Tables A.8 and A.9 report results for a sex-based (men and women) and age-based (under and over 45 years old) decomposition of NTS employment. Our SSIV models indicate that job creation from TS employment growth is concentrated around young men in Brazil. The stronger effects for young men may be linked to a potentially greater responsiveness to local labour demand. One possible explanation is that younger workers face lower mobility or adjustment costs, making them more likely to benefit from local employment expansions. Another possibility is higher relative representation in sectors benefitting more from increased local demand. For instance, the Brazilian construction sector can be particularly prominent as cities face urban expansion, strongly benefitting from increased household and business demand. Demand for particular NTS sectors, some of which have higher relative participation of male workers, may therefore also be a factor at play.

### 5.4 Does location matter?

Finally, we try to understand if microregion size matters. Given the concentration of employment, in particular TS employment in Brazil, a relevant question is whether resuls are purely driven by the usual "winners" of globalisation, that is, the largest cities. Weighing our baseline regression by base period (1995) employment in table A.10, we find that the job multiplier effect nearly quadruples. This suggests that larger microregions tend to have stronger job multipliers. However, in a re-sampling exercise where we remove top and bottom 10% microregions in terms of employment size, we confirm that this not mean that only a handful of microregions (São Paulo and Rio de Janeiro) are responsible for the job multiplier effect identified for Brazil (Table A.11).

### 5.5 Alternative outcomes and robustness checks

As is theoretically understood, the job multiplier captures only a partial reaction of the labour market to an exogenous rise in TS employment. Labour mobility is a key determinant to whether additional household income will, for instance, increase employment in NTS. We therefore estimate

the impact of TS employment growth on other components of the total wage bill, namely average hours worked and real hourly wage in NTS. We find evidence that both are also positively impacted, in particular real wages: for a 10% increase in TS employment, our model suggests a 2.3% increase in average real wages in NTS (Table A.12).

As previously highlighted, RAIS lacks information on informal workers, which could raise issues concerning claims to actual job growth in non-tradable services. To control for this, we make use of Brazilian census data to observe full (formal and informal) employment in 2000 and 2010. Following a long-difference specification similar to that of Moretti (Moretti, 2010), we test for the effect of TS job growth on microregion unemplyoment. The findings (Table A.13) suggest that the growth in employment in TS is strongly associated with a reduction in unemployment, strengthening the idea that what is being captured in the baseline model as NTS job growth is not merely the movement of informal workers who obtain formal contracts or from simple sector reallocation.

Another threat to our specification could be an endogenous SSIV due to lagged shares. As explained, we lag shares, as opposed to use base period shares, to better account for structural change in tradable services across a long panel of data. However, time-varying shares raise the possibility of endogenous re-specialization. To address this, we stress test the SSIV by using longer lags (table A.14). Reassuringly, estimated multipliers remain stable and increase monotonically as shares become more predetermined, inconsistent with endogeneity driven by contemporaneous local shocks. The coefficient increase with longer shares may be informing on less short-term job multiplier effects. In this setting, microregions are exposed to TS growth based on increasingly pre-existing sectorial specialisation and less affected by short-run adjustments to recent shocks, which can dampen the job multiplier.

Finally, we try to emulate our results following a similar identification strategy to that of Avdiu et al. (2022). As we have argued, by relying on an exogenous growth of TS jobs, we want to establish a link between services trade and non-tradable job growth in local labour markets. A key advantage of this approach is the detailed sectorial information on services employment, compared to much scarcer information from services trade statistics. As Bartik and Sotherland (2019) show, the robustness of the shift-share instrument improves with the number of underlying sectors, as a more granular decomposition increases the effective number of shocks and reduces the influence of any single sector-specific disturbance. In this sense, our use of detailed service industries helps approximate a "many shocks" environment, strengthening the quasi-experimental variation

exploited by the SSIV. Still, the model excludes trade statistics and considers intra-national, cross-microregion flows as trade.

In this exercise, at the expense of lower sectorial detail, we exploit the exogenous variation of services exports as the "shift" component of our SSIV. The growth rate of services exports is obtained from the OECD ICIO (Inter-Country Input-Output) tables published in 2025, which contained services trade information for 21 unique services sectors.[12] To proxy for the rise in Brazilian services exports, we follow Autor et al. (2020) and take the growth rate of services exports for a similar basket of Latin American economies.[13] Table A.15 compares baseline results to the ones obtained through a trade-based shit-share instrument. Column 4 suggests a significant and stronger result for the job multiplier, which nearly doubles in size. While magnitudes differ, which is expect for this different exercise, results further consolidate the idea that there is a nexus between services trade and job growth in the non-tradable sector. In sum, as services exports increase and generates job growth in tradable services, positive spillover effects are felt across local labor markets.

# 6 Conclusion

This paper examines whether tradable services generate local employment spillovers and how these compare to those traditionally attributed to manufacturing. Combining Brazilian administrative data with a panel shift-share instrumental variable strategy, we provide causal evidence that employment growth in tradable services leads to substantial job creation in local non-tradable sectors.

Using high-frequency, fine-grained data, we identify tradable services based on their geographic concentration and estimate a local employment multiplier of approximately 1.3. This exceeds the corresponding multiplier for the broader tradable sector and challenges the conventional emphasis on manufacturing as the primary engine of local spillovers. More broadly, the results highlight the growing importance of service-based activities that are capable of serving non-local demand in shaping regional labour market dynamics.

We further investigate the mechanisms underlying these effects. The evidence points to the

[12] OECD ICIO services sectors are reported at the ISIC Rev. 4 2-digit level or, often, at more aggregate levels. For services, such sectors are: 41-43; 45-47; 49; 50; 51; 52; 53; 55-56; 58-60; 61; 62-63; 64-66; 68; 69-75; 77-82; 84; 85; 86-88; 90-93; 94-96; and 97-98. The full list of ISIC Rev. 4 2-digit sectors is reported in table A.1

[13] This is obtained through a weighted average, based on exports, of the following countries available in the OECD ICIO 2025: Argentina, Mexico, Colombia, Chile, Costa Rica and Peru.

consumption channel as the dominant driver: expansions in tradable services employment increase local household demand for non-tradable goods and services. In contrast, input-output linkages between firms appear to play a more limited role. Our findings also challenge the conventional idea that high-skilled labour is the main driver of the job multiplier effect - we do not find statistically significant differences between blue and white collar workers as drivers of job growth in non-tradable services.

The gains from tradable services employment growth are not evenly distributed. Spillovers are stronger for male workers and for individuals at prime working ages, reflecting the sectoral composition of job creation in Brazil. The magnitude of these effects also varies across space, with larger multipliers in bigger and denser local labour markets, consistent with stronger demand linkages in these environments. Complementary results on wages and hours worked indicate that local labour markets adjust along multiple margins, reinforcing the overall importance of these spillovers.

These findings have implications for regional and labour market policy. As economies - but advanced and emerging ones, like Brazil - continue to shift toward service-based production, understanding the spillover effects frpm sectors exposed to non-local demand becomes increasingly important. While policy discussions have traditionally focused on manufacturing, our results suggest that tradable services can also act as significant drivers of local employment growth.

More broadly, although our baseline empirical design does not directly observe trade flows, the results are consistent with the idea that the increasing tradability of services — including through cross-border trade — can have meaningful local labour market implications. This underscores the need to better integrate services into discussions of economic development and spatial inequality across countries and regions. Crucially, this study underscores the importance of tradable services as drivers of local labour market dynamics, providing new evidence relevant for debates on gains from trade in an increasingly interconnected world.

## References


Abras, A., R. K. Almeida, P. Carneiro, and C. H. L. Corseuil (2018, December). Enforcement of labor regulations and job flows: evidence from brazilian cities. *IZA J. Dev. Migr. 8*(1).

Autor, D., D. Dorn, G. H. Hanson, G. Pisano, and P. Shu (2020). Foreign Competition and Domestic Innovation: Evidence from US Patents. *American Economic Review: Insights 2*(3), 357–374.

Autor, D. H., D. Dorn, and G. H. Hanson (2013, October). The china syndrome: Local labor market effects of import competition in the united states. *Am. Econ. Rev. 103*(6), 2121–2168.

Avdiu, B., K. S. Bagavathinathan, R. Chaurey, and G. Nayyar (2022, June). India's services sector growth: The impact of services trade on non-tradable services. *The World Bank*.

Bartik, T. J. (1991). *Who Benefits from State and Local Economic Development Policies?* W.E. Upjohn Institute.

Bartik, T. J. and N. Sotherland (2019, March). Local Job Multipliers in the United States: Variation with Local Characteristics and with High-Tech Shocks. Technical report, W.E. Upjohn Institute.

Beaudry, C. and A. Schiffauerova (2009, March). Who's right, marshall or jacobs? the localization versus urbanization debate. *Res. Policy 38*(2), 318–337.

Benz, Sebastian; Jaax, A. Y.-Y. V. (2022). Shedding light on the drivers of services tradability over two decades. *OECD Trade Policy Papers*.

Berg, J. (2010). Laws or luck?: understanding rising formality in brazil in the 2000s.

Bloom, N., K. Handley, A. Kurmann, and P. Luck (2024, November). The china shock revisited: Job reallocation and industry switching in u.s. labor markets. Technical report, Cambridge, MA.

Borusyak, K., P. Hull, and X. Jaravel (2025, February). A Practical Guide to Shift-Share Instruments. *Journal of Economic Perspectives 39*(1), 181–204.

Calligaris, S., M. Chaves, C. Criscuolo, J. De Lyon, A. Greppi, and O. Pallanch (2024). Exploring the evolution and the state of competition in the eu. *Protecting competition in a changing world, European Commission*.

Cerqua, A. and G. Pellegrini (2020). Loval multipliers at work. *Industrial and Corporate Change 29*(4), 959–977.

Dauth, W., S. Findeisen, and J. Suedekum (2021, January). Adjusting to globalization in germany. *J. Labor Econ. 39*(1), 263–302.

de Blasio, G. and C. Menon (2011). Local effects of manufacturing employment growth in italy. *Giornale degli Economisti e Annali di Economia, Nuova Serie 70*, 101–112.

Dix-Carneiro, R. and B. K. Kovak (2017, October). Trade liberalization and regional dynamics. *Am. Econ. Rev. 107*(10), 2908–2946.

Frocrain, P. and P.-N. Giraud (2019, January). The Evolution of Tradable and Non Tradable Employment: Evidence from France. *Economie et Statistique / Economics and Statistics* (503d), 87–107.

Gerolimetto, M. and S. Magrini (2016, October). A spatial analysis of employment multipliers in the US. *Lett. Spat. Resour. Sci. 9*(3), 277–285.

Goos, M., J. Konings, and M. Vandeweyer (2018, August). Local high-tech job multipliers in europe. *Ind. Corp. Chang. 27*(4), 639–655.

Herman, Peter R.; Oliver, S. (2023). Trade, policy, and economic development in the digital economy. *Journal of Development Economics 164*, 103–135.

Jensen, J. and L. Kletzer (2005). Tradable services: Understanding the scope and impact of services offshoring. *Brookings Trade Forum*, 75–133.

Kazekami, S. (2017, July). Local multipliers, mobility, and agglomeration economies. *Ind. Relat. (Berkeley) 56*(3), 489–513.

Malgouyres, C. (2017, June). The impact of chinese import competition on the local structure of employment and wages: Evidence from france. *J. Reg. Sci. 57*(3), 411–441.

Moretti, E. (2010, May). Local multipliers. *American Economic Review 100*(2), 373–377.

Moretti, E. and P. Thulin (2013, February). Local multipliers and human capital in the united states and sweden. *Ind. Corp. Chang. 22*(1), 339–362.

Naercio A. Menezes-Filho, M.-A. M. (2011). Labor reallocation in response to trade reform. *NBER Working Paper* (17372).

Nguyen, H. and J. Soh (2017). Employment multipliers over the business cycle. *World Bank Policy Research Working Papers* (8105).

Osman, T. and T. Kemeny (2022, January). Local job multipliers revisited. *Journal of Regional Science 62*(1), 150–170.

Van Dijk, J. J. (2018, March). Robustness of econometrically estimated local multipliers across different methods and data. *J. Reg. Sci. 58*(2), 281–294.

Wang, T. and A. Chanda (2018, June). Manufacturing growth and local employment multipliers in china. *J. Comp. Econ. 46*(2), 515–543.

WTO (2019). *World Trade Report 2019: The future of services trade.* World Trade Organizaton, Geneva.

WTO (2023). *Digitally delivered trade.* Handbook on Measuring Digital Trade.

# A Services categories

Table A.1: Categorising services tradable and non-tradable

| Section (A-U) | Division (2-digit) | (1) | (2) | (3) | Category |
|---|---|---|---|---|---|
| Construction (F) | 41. Construction of buildings | 0.22 | No | No | Non-tradable |
| | 42. Civil engineering | 0.3 | No | No | Non-tradable |
| | 43. Specialized construction activities | 0.63 | Yes | No | Non-tradable |
| Wholesale and retail (G) | 45. Wholesale and retail trade and repair of motor vehicles and motorcycles | 0.19 | No | Yes | Non-tradable |
| | 46. Wholesale trade, except of motor vehicles and motorcycles | 0.11 | No | Yes | Non-tradable |
| | 47. Retail trade, except of motor vehicles and motorcycles | 0.03 | No | Yes | Non-tradable |
| Transportation and storage (H) | 49. Land transport and transport via pipelines | 0.12 | No | No | Tradable |
| | 50. Water transport | 11.49 | Yes | Yes | Tradable |
| | 51. Air transport | 6.73 | Yes | Yes | Tradable |
| | 52. Warehousing and support activities for transportation | 0.78 | Yes | Yes | Tradable |
| | 53. Postal and courier activities | 0.23 | No | Yes | Non-tradable |
| Accomodation and food services (I) | 55. Accommodation | 0.9 | Yes | No | Non-tradable |
| | 56. Food and beverage service activities | 0.3 | No | No | Non-tradable |
| Information and communication (J) | 58. Publishing activities | 3.51 | Yes | Yes | Tradable |
| | 59. Motion picture, video and television programme production, sound recording and music publishing activities | 3.42 | Yes | No | Tradable |
| | 60. Programming and broadcasting activities | 0.89 | Yes | Yes | Tradable |
| | 61. Telecommunications | 0.22 | No | No | Tradable |
| | 62. Computer programming, consultancy and related activities | 3.1 | Yes | Yes | Tradable |
| | 63. Information service activities | 3.76 | Yes | Yes | Tradable |
| Finance and insurance (K) | 64. Financial service activities, except insurance and pension funding | 2.05 | Yes | No | Tradable |
| | 65. Insurance, reinsurance and pension funding, except compulsory social security | 4.9 | Yes | Yes | Tradable |
| | 66. Activities auxiliary to financial service and insurance activities | 9.48 | Yes | Yes | Tradable |
| Real estate (L) | 68. Real estate activities | 0.25 | No | No | Tradable |
| Professional, scientific and technical activities (M) | 69. Legal and accounting activities | 0.07 | No | No | Tradable |
| | 70. Activities of head offices; management consultancy activities | 6.62 | Yes | Yes | Tradable |
| | 71. Architectural and engineering activities; technical testing and analysis | 0.61 | Yes | Yes | Tradable |
| | 72. Scientific research and development | 2.72 | Yes | Yes | Tradable |

*Continued on next page*

| *Continued from previous page* | | | | | |
|---|---|---|---|---|---|
| Section (A-U) | Division (2-digit) | (1) | (2) | (3) | Category |
| | 73. Advertising and market research | 3.26 | Yes | Yes | Tradable |
| | 74. Other professional, scientific and technical activities | 4.08 | Yes | No | Tradable |
| | 75. Veterinary activities | 0.22 | No | No | Non-tradable |
| Administrative and support services (N) | 77. Rental and leasing activities | 0.79 | Yes | No | Non-tradable |
| | 78. Employment activities | 4.01 | Yes | Yes | Tradable |
| | 79. Travel agency, tour operator, reservation service and related activities | 1.79 | Yes | Yes | Tradable |
| | 80. Security and investigation activities | 1.3 | Yes | No | Non-tradable |
| | 81. Services to buildings and landscape activities | 1.71 | Yes | No | Non-tradable |
| | 82. Office administrative, office support and other business support activities | 1.07 | Yes | Yes | Tradable |
| Public administration and defence (O) | 84. Public administration and defence; compulsory social security | 0.96 | Yes | No | Non-tradable |
| Education (P) | 85. Education | 0.2 | No | No | Non-tradable |
| Human health and social work activities (Q) | 86. Human health activities | 0.18 | No | Yes | Non-tradable |
| | 87. Residential care activities | 0.39 | No | No | Non-tradable |
| | 88. Social work activities without accommodation | 2.35 | Yes | No | Non-tradable |
| Arts, entertainment and recreation (R) | 90. Creative, arts and entertainment activities | 1.53 | Yes | No | Non-tradable |
| | 91. Libraries, archives, museums and other cultural activities | 2.54 | Yes | No | Non-tradable |
| | 92. Gambling and betting activities | 2.91 | Yes | No | Non-tradable |
| | 93. Sports activities, amusement and recreation activities | 0.28 | No | No | Non-tradable |
| Other service activities (S) | 94. Activities of membership organizations | 0.39 | No | Yes | Non-tradable |
| | 95. Repair of computers and personal and household goods | 0.55 | Yes | No | Non-tradable |
| | 96. Other personal service activities | 0.15 | No | No | Non-tradable |
| Households activities (T) | 97. Activities of households as employers of domestic personnel | 3.42 | Yes | No | Non-tradable |
| | 98. Undifferentiated goods- and services-producing activities of private households for own use | Not included in RAIS | | | Non-tradable |
| Extraterritorial bodies (U) | 99. Activities of extraterritorial organizations and bodies | 43.3 | Yes | No | Non-tradable |

*Notes:* Column (1) reports the geographic concentration of the sector in 2023, following equation 1. Column (2) indicates whether the sector is above the tradability cutoff (lowest geographic concentration in manufacturing). This cutoff in 2023 refers to division 33 (Repair and installation of machinery and equipment), which reported a geographic concentration of 0.44. Column (3) indicates whether the sector is consistently above or below the cutoff for every year between 1994 and 2023.

Table A.2: Composition of tradable services in Brazil by shares of 2-digit sector employment, 1995 and 2023

| Section (A-U) | Division (2-digit) | 1995 | 2023 | Difference |
|---|---|---|---|---|
| Transportation and storage (H) | 49. Land transport and transport via pipelines | 27.5% | 20.2% | -7.3 |
| | 50. Water transport | 1.0% | 0.5% | -0.5 |
| | 51. Air transport | 1.3% | 0.6% | -0.7 |
| | 52. Warehousing and support activities for transportation | 2.5% | 5.3% | 2.8 |
| Information and communication (J) | 58. Publishing activities | 1.7% | 2.4% | 0.7 |
| | 59. Motion picture, video and television programme production, sound recording and music publishing activities | 2.8% | 0.3% | -2.5 |
| | 60. Programming and broadcasting activities | 1.0% | 0.8% | -0.2 |
| | 61. Telecommunications | 3.5% | 2.9% | -0.6 |
| | 62. Computer programming, consultancy and related activities | 0.9% | 4.7% | 3.8 |
| | 63. Information service activities | 1.8% | 1.4% | -0.4 |
| Finance and insurance (K) | 64. Financial service activities, except insurance and pension funding | 17.7% | 7.3% | -10.4 |
| | 65. Insurance, reinsurance and pension funding, except compulsory social security | 1.7% | 2.0% | 0.3 |
| | 66. Activities auxiliary to financial service and insurance activities | 1.3% | 1.9% | 0.6 |
| Real estate (L) | 68. Real estate activities | 9.8% | 2.1% | -7.7 |
| Professional, scientific and technical activities (M) | 69. Legal and accounting activities | 4.0% | 5.6% | 1.6 |
| | 70. Activities of head offices; management consultancy activities | 3.2% | 1.9% | -1.3 |
| | 71. Architectural and engineering activities; technical testing and analysis | 1.3% | 4.3% | 3.0 |
| | 72. Scientific research and development | 1.3% | 0.6% | -0.7 |
| | 73. Advertising and market research | 0.9% | 1.9% | 1.0 |
| | 74. Other professional, scientific and technical activities | 0.4% | 1.7% | 1.3 |
| Administrative and support services (N) | 78. Employment activities | 4.0% | 9.4% | 5.4 |
| | 79. Travel agency, tour operator, reservation service and related activities | 3.2% | 1.9% | -1.9 |
| | 82. Office administrative, office support and other business support activities | 9.2% | 21.5% | 12.3 |

*Notes:* Columns 1995 and 2023 report the the given sector (division)'s employment share of tradable services for each respective year. The last column (difference) reports percentage point differences between both years.

Table A.3: Categorising market non-tradable services into business-to-business- and business-to-consumer-intensive

| Section (A-U) | Division (2-digit) | (1) | Category |
|---|---|---|---|
| Transportation and storage (H) | 53. Postal and courier activities | 92.4% | B2B |
| Accomodation and food services (I) | 55. Accommodation<br>56. Food and beverage service activities | 18.3% | B2C |
| Professional, scientific and technical activities (M) | 75. Veterinary activities | - | B2C |
| Administrative and support services (N) | 77. Rental and leasing activities<br>80. Security and investigation activities<br>81. Services to buildings and landscape activities | 87.7% | B2B |
| Education (P) | 85. Education | 0.9% | B2C |
| Human health and social work activities (Q) | 86. Human health activities<br>87. Residential care activities<br>88. Social work activities without accommodation | 1.7% | B2C |
| Arts, entertainment and recreation (R) | 90. Creative, arts and entertainment activities<br>91. Libraries, archives, museums and other cultural activities<br>92. Gambling and betting activities<br>93. Sports activities, amusement and recreation activities | 7.5% | B2C |
| Other service activities (S) | 94. Activities of membership organizations<br>95. Repair of computers and personal and household goods<br>96. Other personal service activities | 9.7% | B2C |
| Households activities (T) | 97. Activities of households as employers of domestic personnel<br>98. Undifferentiated goods- and services-producing activities of private households for own use | 0% | B2C |

*Notes:* Column (1) reports the share of output (excluding gross fixed capital formation) of the sector that is destined to intermediate consumption of other firms. Results are estimated using the OECD Inter Country Input-Output tables, which contain less sector detail than RAIS. Since results are only available at the section level, and that veterinary activities are bundled within a largely tradable B2B-intensive section, results are not reported for this sector. GIven its description in ISIC Rev. 4, it is conceptually presumed to a B2C-intensive sector.

# B Additional results

Table A.4: Alternative specifications: base period controls

| | *emp.NTS* (without geographic FE) | | | | *emp.NTS* (with state FE) | | | |
|---|---|---|---|---|---|---|---|---|
| | (1) OLS | (2) SSIV | (3) OLS | (4) SSIV | (5) OLS | (6) SSIV | (7) OLS | (8) SSIV |
| *emp.TS* | 0.649*** | 0.529*** | 0.339*** | 0.325*** | 0.713*** | 0.574*** | 0.345*** | 0.309*** |
| | (0.014) | (0.025) | (0.020) | (0.066) | (0.012) | (0.034) | (0.021) | (0.069) |
| 95.*sh.eduhigh* | | | 1.700*** | 1.750*** | | | 1.480*** | 1.540*** |
| | | | (0.242) | (0.273) | | | (0.284) | (0.317) |
| 95.*sh.mfgemp* | | | -0.541*** | -0.554*** | | | -0.535*** | -0.559*** |
| | | | (0.111) | (0.125) | | | (0.096) | (0.109) |
| 95.*sh.fememp* | | | 0.670*** | 0.635** | | | 0.417** | 0.374* |
| | | | (0.123) | (0.198) | | | (0.146) | (0.168) |
| 95.*logemp* | | | 0.317*** | 0.329*** | | | 0.393*** | 0.427*** |
| | | | (0.026) | (0.066) | | | (0.030) | (0.073) |
| 91.*logpop* | | | 0.222*** | 0.225*** | | | 0.139*** | 0.147*** |
| | | | (0.027) | (0.028) | | | (0.025) | (0.029) |
| 91.*sh.infemp* | | | 0.397** | 0.389** | | | 0.325* | 0.326* |
| | | | (0.137) | (0.132) | | | (0.139) | (0.145) |
| F-stat | – | 697.49 | – | 313.36 | – | 421.34 | – | 360.13 |
| Adj. $R^2$ | 0.866 | 0.841 | 0.940 | 0.940 | 0.912 | 0.889 | 0.950 | 0.950 |
| Observations | 14,790 | 14,790 | 13,572 | 13,572 | 14,790 | 14,790 | 13,572 | 13,572 |
| FEs | yr | yr | yr | yr | yr, geo (state) | yr, geo (state) | yr, geo (state) | yr, geo (state) |

Standard errors clustered at the GEO level. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Samples with full set of controls are affected, since not all Brazilians microregions are accurately identifiable in the Brazilian 1991 Census. Variables for population and share of informal employment are taken from this source.

Table A.5: Alternative specifications: moving controls

| | *emp.NTS* (baseline) | | *emp.NTS* (with controls) | |
|---|---|---|---|---|
| | (1) OLS | (2) SSIV | (3) OLS | (4) SSIV |
| *emp.TS* | 0.202*** | 0.294* | 0.201*** | 0.298** |
| | (0.031) | (0.120) | (0.027) | (0.107) |
| *sh.eduhigh* | | | 2.17*** | 2.12*** |
| | | | (0.276) | (0.283) |
| *sh.mfgemp* | | | -0.682*** | -0.621** |
| | | | (0.180) | (0.194) |
| *sh.fememp* | | | 1.43*** | 1.60*** |
| | | | (0.366) | (0.398) |
| F-stat | – | 314.49 | – | 205.78 |
| Adj. $R^2$ | 0.967 | 0.966 | 0.974 | 0.973 |
| Observations | 14790 | 14790 | 14790 | 14790 |
| FEs | geo, yr | geo, yr | geo, yr | geo, yr |

Standard errors clustered at the GEO level. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table A.6: Heterogeneity: skills

| | *emp.NTS* | | | |
|---|---|---|---|---|
| | (1) OLS | (2) SSIV | (3) OLS | (4) SSIV |
| High-skilled (*emp.TS*) | 0.248*** | 0.187** | | |
| | (0.017) | (0.083) | | |
| Low-skilled (*emp.TS*) | | | 0.055*** | 0.202*** |
| | | | (0.017) | (0.060) |
| F-stat | – | 120.85 | – | 264.59 |
| Adj. $R^2$ | 0.970 | 0.970 | 0.965 | 0.962 |
| Observations | 14790 | 14790 | 14790 | 14790 |
| FEs | geo, yr | geo, yr | geo, yr | geo, yr |

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. SEs clustered at microregion level.

Table A.7: Mechanisms: sectorial split of non-tradable services

| | B2B-intensive (*emp.NTS*) | | B2C-intensive (*emp.NTS*) | |
|---|---|---|---|---|
| | (1) OLS | (2) SSIV | (3) OLS | (4) SSIV |
| *emp.TS* | 1.038*** | 0.212 | 0.902*** | 0.356*** |
| | (0.004) | (0.149) | (0.002) | (0.108) |
| F-stat | – | 314.49 | – | 314.49 |
| Adj. $R^2$ | 0.853 | 0.940 | 0.899 | 0.974 |
| Observations | 14790 | 14790 | 14790 | 14790 |
| FEs | geo, yr | geo, yr | geo, yr | geo, yr |

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. SEs clustered at microregion level.

Table A.8: Heterogeneity: sex

| | Men ($emp.NTS$) | | Women ($emp.NTS$) | |
|---|---|---|---|---|
| | (1) OLS | (2) SSIV | (3) OLS | (4) SSIV |
| $emp.TS$ | 0.255*** | 0.513*** | 0.160*** | 0.121 |
| | (0.029) | (0.130) | (0.036) | (0.122) |
| F-stat | – | 314.49 | – | 314.49 |
| Adj. $R^2$ | 0.968 | 0.964 | 0.959 | 0.959 |
| Observations | 14790 | 14790 | 14790 | 14790 |
| FEs | geo, yr | geo, yr | geo, yr | geo, yr |

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. SEs clustered at microregion level.

Table A.9: Heterogeneity: age

| | Under-45 ($emp.NTS$) | | 45+ ($emp.NTS$) | |
|---|---|---|---|---|
| | (1) OLS | (2) SSIV | (3) OLS | (4) SSIV |
| $emp.TS$ | 0.219*** | 0.358*** | 0.174*** | 0.071 |
| | (0.031) | (0.125) | (0.034) | (0.103) |
| F-stat | – | 314.49 | – | 314.49 |
| Adj. $R^2$ | 0.965 | 0.964 | 0.967 | 0.966 |
| Observations | 14790 | 14790 | 14790 | 14790 |
| FEs | geo, yr | geo, yr | geo, yr | geo, yr |

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. SEs clustered at microregion level.

Table A.10: Weighted regression by pre-period microregion employment

| | $emp.NTS$ | | | |
|---|---|---|---|---|
| | (1) OLS | (2) SSIV | (3) OLS | (4) SSIV |
| $emp.TS$ | 0.202*** | 0.294** | 0.196*** | 1.348*** |
| | (0.031) | (0.120) | (0.024) | (0.002) |
| F-stat | – | 314.49 | – | – |
| Adj. $R^2$ | 0.967 | 0.966 | 0.978 | 0.004 |
| Observations | 14790 | 14790 | 14790 | 14790 |
| FEs | geo, yr | geo, yr | geo, yr | geo, yr |

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. SEs clustered at microregion level. Models (3) and (4) are weighted on the log of pre-period employment.

Table A.11: Baseline model, without bottom and top decile of microregions (by employment)

| | $emp.NTS$ | | | |
|---|---|---|---|---|
| | (1) OLS | (2) SSIV | (3) OLS | (4) SSIV |
| $emp.TS$ | 0.220*** | 0.249** | | |
| | (0.030) | (0.097) | | |
| $emp.T$ | | | 0.147*** | 0.585*** |
| | | | (0.042) | (0.196) |
| F-stat | – | 101.86 | – | 22.83 |
| Adj. $R^2$ | 0.940 | 0.940 | 0.922 | 0.886 |
| Observations | 9454 | 9454 | 7540 | 7540 |
| FEs | geo, yr | geo, yr | geo, yr | geo, yr |

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. SEs clustered at microregion level.

Table A.12: Beyond employment: additional impacts of growth in employment in tradable services

| | Log hours per NTS worker | | Log real NTS hourly wage | |
|---|---|---|---|---|
| | (1) OLS | (2) SSIV | (3) OLS | (4) SSIV |
| *emp.TS* | 0.016*** | 0.054*** | 0.068*** | 0.235*** |
| | (0.003) | (0.019) | (0.010) | (0.049) |
| F-stat | – | 148.24 | – | 148.24 |
| Adj. $R^2$ | 0.618 | 0.570 | 0.866 | 0.835 |
| Observations | 12240 | 12240 | 12240 | 12240 |
| FEs | geo, yr | geo, yr | geo, yr | geo, yr |

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. SEs clustered at microregion level.

Table A.13: Including the informal sector: the impact of services trade on unemployment (2000 and 2010)

| | $\Delta emp.NTS$ | | Unemployment rate | |
|---|---|---|---|---|
| | (1) OLS | (2) SSIV | (3) OLS | (4) SSIV |
| $\Delta emp.TS$ | 0.826*** | 1.204*** | -4.644*** | -8.165*** |
| | (0.034) | (0.049) | (0.265) | (0.407) |
| F-stat | – | 26.18 | – | 22.46 |
| Adj. $R^2$ | -1.632 | -2.012 | -1.432 | -2.435 |
| Observations | 510 | 509 | 473 | 472 |
| FEs | | | | |

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. SEs clustered at microregion level.

Table A.14: Lagged shares

| | *emp.NTS* | | | |
|---|---|---|---|---|
| | (1) OLS | (2) SSIV (baseline) | (3) SSIV (2-y lag) | (4) SSIV (3-y lag) |
| *emp.TS* | 0.202*** | 0.294** | 0.458*** | 0.606*** |
| | (0.031) | (0.120) | (0.146) | (0.168) |
| F-stat | – | 314.49 | 216.31 | 227.53 |
| Adj. $R^2$ | 0.967 | 0.966 | 0.967 | 0.967 |
| Observations | 14790 | 14790 | 14790 | 14790 |
| FEs | geo, yr | geo, yr | geo, yr | geo, yr |

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. SEs clustered at microregion level.

Models (3) and (4) do not have a reduced sample. This implies that lags are smaller for earlier years to keep sample consistency. For instance, 1995 is included in both years with a one-year lag.

Table A.15: Alternative instrument: trade-based shock

| | *emp.NTS* (baseline) | | *emp.NTS* (trade-based shock) | |
|---|---|---|---|---|
| | (1) OLS | (2) SSIV | (3) OLS | (4) SSIV |
| *emp.TS* | 0.202*** | 0.294** | 0.179*** | 0.521*** |
| | (0.031) | (0.120) | (0.030) | (0.120) |
| F-stat | – | 314.49 | – | 188.84 |
| Adj. $R^2$ | 0.967 | 0.966 | 0.970 | 0.964 |
| Observations | 14790 | 14790 | 13770 | 13770 |
| FEs | geo, yr | geo, yr | geo, yr | geo, yr |

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. SEs clustered at microregion level.

Columns 3-4 are based on a trade-based IV, which uses data only available from 1995 to 2022. Therefore, the sample is smaller due to the loss of information for 1995 and 2023.